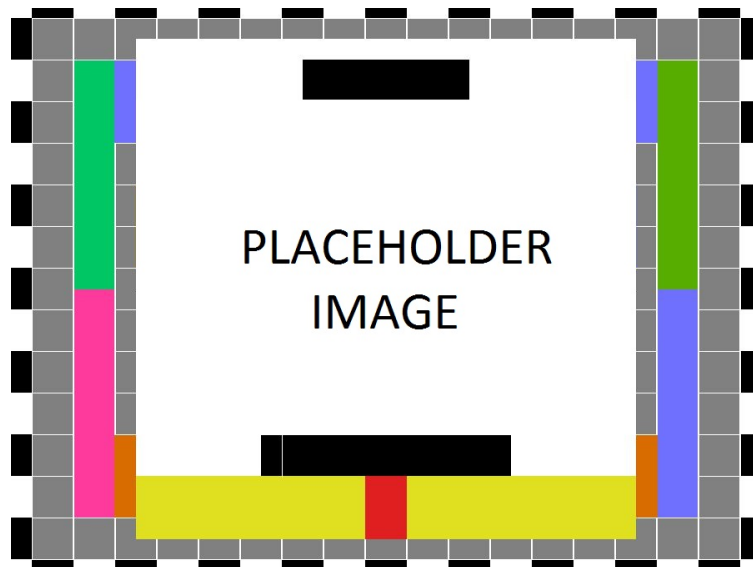


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

**ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN**

ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.02.2026	GH	document creation
1.0	09.15.2026	AT, GH, CB	official release

CONTENTS

1	Problem Statement	5
2	Methodology	5
3	Value Proposition	5
4	Development Milestones	5
5	Background	6
6	Related Work	6
7	System Overview	6
8	Project Management Approach	6
9	Roles & Responsibilities	6
10	Cost Proposal	6
11	Facilities & Equipment	6
12	Assumptions	6
13	Constraints	7
14	Risks	7
15	Documentation & Reporting	7

1 PROBLEM STATEMENT

Residents report non-emergency issues — potholes, broken streetlights, illegal dumping — through phone calls and email that land in scattered inboxes. Public works staff cannot see the full backlog, duplicates pile up, and residents get no status back, eroding trust. The city needs a single place for residents to report issues and for staff to see, prioritize, and resolve them, with visible status.

2 METHODOLOGY

Team Fullstack will build CivicPulse Community Service-Request Dashboard, a web application with a public resident portal for submitting and tracking geolocated requests and a staff dashboard for triage, assignment, and status updates. In short: we are building a website where residents report neighborhood issues and city staff manage them to resolution on a shared dashboard.

3 VALUE PROPOSITION

For the city, CivicPulse consolidates intake, removes duplicate handling, and gives residents self-service status, reducing call volume and improving accountability. For the university, the project exercises a complete secure multi-tier web application — authentication, role-based access, geospatial data, and reporting.

4 DEVELOPMENT MILESTONES

- Project Charter first draft - September 2026
- System Requirements Specification - October 2026
- Architectural Design Specification - November 2026
- Demonstration of request submission & map view - December 2026
- Detailed Design Specification - February 2027
- Demonstration of staff triage & assignment - March 2027
- Demonstration of status notifications - April 2027
- CoE Innovation Day poster presentation - April 2027
- Final Project Demonstration (end-to-end workflow) - May 2027

5 BACKGROUND

The department tracks requests in spreadsheets and email, with no resident-facing status. They asked the CSE program for a modern web front door integrated with their existing single sign-on. No prior relationship; requirements from two interviews and a review of their current intake spreadsheet.

6 RELATED WORK

Commercial 311/CRM platforms (e.g. SeeClickFix-style products) exist [1] but are subscription-priced and not tailored to the city's SSO and categories. CivicPulse targets a lightweight, self-hostable alternative. (Add ≥ 5 IEEE references in the final version.)

7 SYSTEM OVERVIEW

A responsive web frontend (resident portal + staff dashboard) talks to a backend API that enforces authentication and role-based access, persists requests with location in a database, and sends status notifications by email. A map component visualizes open requests. See the ADS for the three-tier design.

8 PROJECT MANAGEMENT APPROACH

Iterative Scrum [1, 2] per ISO/IEC/IEEE 16326:2019 [3] within ISO/IEC/IEEE 12207:2017 [4]. Two-week sprints; estimation in task-hours; progress by burndown and by requirement/test coverage. Trunk-based Git with CI; tagged document releases.

9 ROLES & RESPONSIBILITIES

Sponsor contact: public-works IT liaison. Product Owner: Grace Hopper. Scrum Master: Alan Turing. Frontend: Ada Lovelace; Backend/API: John Von Neumann; Data/geo: Charles Babbage; Auth/notifications: Grace Hopper; Integration/test: Alan Turing.

10 COST PROPOSAL

Mainly cloud hosting for a staging environment plus a mapping/geocoding API (free tier where possible); no custom hardware. Funded by the default CSE stipend; largest line item is hosting.

11 FACILITIES & EQUIPMENT

Senior Design lab workstations; a cloud project for staging; access to a sponsor test SSO realm by the 3rd sprint.

12 ASSUMPTIONS

- A sponsor SSO/OIDC test realm will be available by the 3rd sprint.
- The city will provide the list of request categories and routing rules.
- A free-tier geocoding/mapping API will meet demo volume.
- Residents access the portal from modern mobile and desktop browsers.
- Email delivery via a provided SMTP relay.

13 CONSTRAINTS

- Final demonstration must be completed by May 1st, 2027.
- Total development cost must not exceed \$800.
- The application must meet WCAG 2.2 AA accessibility.
- Resident personal data handling must follow the city's privacy policy.
- Staff functions must be restricted by role (no public access).

14 RISKS

Risk description	Probability	Loss (days)	Exposure (days)
SSO/OIDC integration with city realm delayed	0.40	12	4.8
Scope creep in staff dashboard features	0.45	10	4.5
Geocoding API limits/costs on real volume	0.30	8	2.4
Accessibility rework late in the project	0.35	7	2.5
Email deliverability issues	0.20	6	1.2

Table 2: Highest-exposure project risks

15 DOCUMENTATION & REPORTING

Documentation deliverables in Git, released at their planning sprints; recurring sprint artifacts and engineering notebooks each sprint. Closeout: deployed application, poster, public web page, demo video, documented source (MIT), infrastructure-as-code deployment scripts, and admin/user manuals.

REFERENCES

- [1] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*, 1st ed. Addison-Wesley Professional, 2012.
- [2] K. Schwaber and J. Sutherland, "The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game," 2020.
- [3] ISO/IEC/IEEE, "ISO/IEC/IEEE 16326:2019 — Systems and software engineering — Life cycle processes — Project management," International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2019.
- [4] —, "ISO/IEC/IEEE 12207:2017 — Systems and software engineering — Software life cycle processes," International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2017.